

Learning from Weather Forecasts and Short-Run Adaptation: Evidence from an At-Scale Experiment

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Abstract

Can low-cost digital forecasts help farmers adapt to weather risk? We study a randomized saturation experiment with 400,000 cotton growers in Punjab, Pakistan. Treated farmers received localized two-day ahead forecasts; saturation varied across tehsils to measure direct effects and spillovers. Forecasts increased weather-information sharing and shifted production timing. Effects are clearest in high-saturation tehsils: treated and untreated farmers were more likely to avoid rain when irrigating and applying fertilizer or pesticides, with suggestive evidence of information diffusion. Heat avoidance is weaker. Yields are positive but imprecise; while input spending rises significantly.

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1 Introduction

Weather risk is a central constraint on agricultural production, especially for small-scale farmers in low- and middle-income countries (Lesk et al., 2016; Wollburg et al., 2024). Short-range forecasts have the potential to help farmers adjust the timing of irrigation, fertilizer, pesticides, and other field activities, and they can often be delivered at scale at little or no cost to recipients (Rosenzweig and Udry, 2019; Molina and Rudik, 2023; Shrader et al., 2023). Yet many farmers still lack access to localized forecasts that are relevant for production decisions (Linsenmeier and Shrader, 2023). This paper studies whether phone-based, geo-localized weather forecasts change farmers’ short-run production choices. We also study whether forecast information diffuses beyond directly treated farmers, by potentially generating spillovers.

In collaboration with Precision Development (PxD) and the forecast provider CFAN, we implemented the experiment through an existing phone-based advisory service for cotton growers in Punjab, Pakistan. The intervention was motivated by prior PxD research showing that, before the experiment, only 45% of cotton growers reported consistent access to weather information, typically through radio or television.¹ Moreover, available forecasts were generally provided at a coarser geographic level (e.g., district instead of tehsil level). The experimental frame consisted of approximately 400,000 listed phone numbers across 35 tehsils. Farmers assigned to treatment received automated voice messages with localized two-day forecasts, including expected rainfall, maximum temperature, and minimum temperature, in addition to the standard agronomic advisory messages received by all farmers.

The randomization had two stages. First, we randomly assigned tehsils to treatment saturation levels, which determined the share of listed farmers in each tehsil assigned to receive weather forecasts. Second, within each tehsil, we randomly assigned individual phone numbers either to a control group that continued receiving standard agronomic advisory messages or to a treatment group that received the same advisory messages plus localized two-day-ahead weather forecasts. We interpret the estimates as intention-to-treat effects of providing geo-localized forecasts in an environment where farmers may also have access to other, less localized sources of weather information.

This design allows us to estimate the direct effect of being assigned access to forecasts, as well as spillover effects from living in a tehsil where a larger share of farmers was assigned to receive

¹See <https://precisiondev.org/weather-forecasting-product-for-punjab-pakistan/>.

forecasts. Because weather information may diffuse across farmers, the randomized saturation design is central to the empirical analysis: by varying both individual assignment and the share of treated farmers within each tehsil, we can separately estimate direct effects and spillover effects. This is motivated by the potential of spillovers, as about 87% of farmers reported sharing weather information with other farmers.

To measure how farmers responded during the growing season, we conducted high-frequency phone surveys with up to 400 farmers per week. These surveys collected real-time information on weather beliefs, forecast receipt and sharing, and the timing of agricultural activities, including irrigation, fertilizer application, and pesticide use. We then link reported activity dates to realized weather and forecasts to test whether farmers shifted input use away from rain and extreme heat. Finally, we conducted an endline survey after the season to measure yields, input expenditures, revenues, and profits.

We first examine whether the forecasts changed farmers' weather beliefs. We measure belief alignment by comparing farmers' next-day weather predictions with the forecasts delivered by the program. For rainfall, directly treated farmers are 3.4 percentage points more likely to make predictions that align with the forecast. We also find evidence of spillovers: a 10 percentage point increase in the tehsil treatment share increases forecast alignment by about 1 percentage point. Results for maximum and minimum temperature point in the same direction, with treated farmers reporting predictions closer to the forecast and larger estimated effects of tehsil saturation. In saturated specifications, the largest belief changes appear among treated farmers in high-saturation tehsils, although these decomposed estimates are less precise. Overall, the belief results suggest that farmers attended to the forecasts and that information diffusion amplified the effects of direct treatment.

The forecasts also changed information flows among farmers. About 40% of control farmers reported sharing weather information with peers. Assignment to treatment increased this probability by 10–13 percentage points. Consistent with information diffusion, untreated farmers in high-saturation tehsils were also more likely to receive weather information from peers than untreated farmers in low-saturation tehsils. These patterns suggest that peer-to-peer communication is a plausible channel through which forecast information reached farmers who were not directly treated.

We next study whether forecasts changed short-run production decisions. Using the dates farmers reported irrigating and applying fertilizer and pesticides, we test whether these activities

were shifted away from days with rain. This margin is economically relevant because applying inputs before or during rainfall can reduce input effectiveness and increase runoff. The largest effects occur in high-saturation tehsils. Relative to control farmers in low-saturation tehsils, directly treated farmers in high-saturation tehsils are 40–68% more likely to avoid rain when irrigating and applying fertilizer or pesticides. Untreated farmers in high-saturation tehsils are also 24–47% more likely to avoid rain, consistent with spillovers in behavior. Treated farmers in low-saturation tehsils move in the same direction, but the estimates are smaller and less precise. Patterns for avoiding extreme heat are similar in sign but statistically weaker, suggesting that forecasts affected some heat-related decisions, though less robustly than rain-related input timing.

Endline outcomes suggest more ambiguous effects on agricultural productivity and profits. Yield estimates are positive for treated farmers, including in high-saturation tehsils, but most estimates are imprecise and only marginally significant for treated farmers in low-saturation tehsils. Profit and revenue effects are less clear. Input expenditures increase, especially in higher-saturation tehsils, and there is suggestive evidence that farmers adjusted harvest timing and received lower output prices, although these estimates are noisy. These patterns suggest that forecasts changed economically meaningful production decisions, but that short-run behavioral adjustment did not translate unambiguously into higher profits potentially due to lower power on profits. Taken together, the results show that localized forecasts can affect farmers’ beliefs, information sharing, and input timing, while also highlighting that the welfare effects of short-run adaptation depend on adjustment costs, market timing, and spillovers.

This paper contributes to three related literatures. First, we build on the agricultural development literature showing that farmers learn from social networks and that information sharing is pervasive in rural settings ([Foster and Rosenzweig, 1995](#); [Munshi, 2004](#); [Bandiera and Rasul, 2006](#)). Second, we contribute to a growing literature on weather forecasts and agricultural decision-making. Prior work shows that farmers can learn to interpret forecasts with limited training ([Barnett-Howell, 2021](#); [Ahmad et al., 2022](#); [Cole et al., 2023](#)) and that weather forecasts can affect agricultural investment and welfare-relevant outcomes ([Rosenzweig and Udry, 2019](#); [Burlig et al., 2024](#)). Relative to this work, our focus is on short-range forecasts and short-run adaptation. Because forecasts are delivered with a two-day lead time and we collect high-frequency survey data during the agricultural season, we can directly measure whether farmers adjust the timing of irrigation, fertilizer, and pesticide application around realized rain and heat. Third, we contribute to the literature on digital agricultural extension and scalable information services ([Fafchamps and Minten, 2012](#); [Fosu](#)

et al., 2018; Camacho and Conover, 2019; Yegbemey et al., 2023; Fabregas et al., 2019; Ferdinand et al., 2021; Fabregas et al., 2024). Our design adds to this literature by evaluating an at-scale forecasting service with randomized variation in treatment saturation, allowing us to study both direct effects and spillovers.

We also connect to the literature on saturation designs and inference under interference (Hudgens and Halloran, 2008; Baird et al., 2018; Vazquez-Bare, 2023). Saturation designs are especially useful for information interventions, where untreated individuals may be indirectly exposed through treated peers. Such spillovers have been studied in several at-scale information experiments, including tax compliance, political advertising, and labor-market interventions (Cruces et al., 2024; Gerber et al., 2011; Enríquez et al., 2024; Duflo and Saez, 2003; Crépon et al., 2013).

Finally, our companion work, Viviano and Rudder (2024), develops methods for using local perturbations in saturation rates to estimate marginal policy effects and study whether spillovers level off when only a finite number of large clusters is available. It focuses on the analysis of effects on beliefs for one particular outcome (temperature forecasts) to illustrate the methodological advances of our new experimental design. In this paper, instead, we study how effects on beliefs may affect behavior through direct and indirect effects: we use the same at-scale saturation experiment to provide a comprehensive empirical analysis of how localized forecasts affect farmers’ information, peer-to-peer diffusion, short-run production behavior, and end-of-season outcomes.

2 Background

Cotton is a politically important, high-value cash crop that contributes to Pakistan’s domestic textile manufacturing. All of cotton production is sold through local intermediaries into Pakistan’s \$17 billion textile industry which accounts for 60% of export values.² Cotton growing is concentrated in the Indus River valley in Punjab and Sindh provinces. The majority of production (66%) is in Punjab province, concentrated in 11 out of 23 districts (USDA, 2023).³ Cotton production is input intensive and grown under irrigation. Farmers use both government-supported irrigation canals and privately-operated tubewell systems. High quality weather forecasts enable farmers to plan important farming tasks - planting, irrigation dates, fertilizer, pesticides, herbicides, and fungicides.

In Punjab province, 1.6 million farmers grow cotton and the majority have farms that are

²Source: World Bank’s World Integrated Trade Solution database. <https://wits.worldbank.org/>

³USDA Foreign Agriculture Service. <https://ipad.fas.usda.gov/highlights/2023/08/Pakistan/index.pdf>

less than 5 acres.⁴ Because it is difficult to reach large numbers of farmers through traditional in-person agricultural extension, our NGO collaborator Precision Development (PxD) launched an agronomic advisory service for cotton growers in 2021. The program included a roster of 400,000 farmer phone-numbers located in 40 cotton-growing tehsils in Punjab Province, representing nearly one-third of all cotton growers in the province.⁵ The roster of farmer phone numbers came from the Punjab Department of Agriculture and consisted of cotton farmers who signed up for government agricultural services. For the 2022 cotton growing season, the NGO expanded operations by adding a weather forecasting service. Both agronomic advisory and weather forecasts were delivered to farmers via automated voice calls. The phone number that appeared on calls was known by farmers to indicate message delivery from the Punjab Department of Agriculture. Farmers could opt out of messages by blocking the phone number or unsubscribing, although in practice this was not common.

Before the intervention, farmers could access weather information from radio, television, smart-phone applications, and the Pakistan Meteorological Department. However, these sources did not provide systematic, phone-based delivery of short-range, localized forecasts directly to farmers. Pilot surveys suggest that access remained incomplete and that demand for weather information was high: about half of farmers reported having access to weather information, while 87% reported sharing weather information with others in their village. These patterns also suggest that forecast information could diffuse beyond directly treated farmers. More broadly, farmers in this setting commonly rely on social and institutional sources when adopting new agricultural practices. Among farmers who reported adopting a new practice in the prior year, 30% reported learning from peers, 47% from government extension officers, and 26% from agrodealers. These facts motivate both the intervention, which delivered localized forecasts directly by phone, and the saturation design, which allows us to measure spillovers from information sharing.

3 Research Design and Data

3.1 Randomization

We implemented a two-wave randomized saturation design at the tehsil level. Out of 40 eligible tehsils, five were set aside by the implementing partner for a separate pilot, leaving 35 tehsils in

⁴Agricultural Census by the Pakistan Bureau of Statistics.

⁵Tehsils are the second smallest administrative unit in Pakistan, equivalent to a county in the United States.

the experimental sample. The sample frame consisted of approximately 400,000 existing users of PxD’s digital cotton advisory service. Tehsils were assigned to low-, medium-, and high-saturation categories, stratified by district. Within each tehsil, individual phone numbers were then randomly assigned either to continue receiving PxD’s standard cotton advisory calls or to receive the same advisory calls plus two-day localized weather forecasts. We treat tehsils as clusters because they were the most granular reliable location available in the government roster and because they are a relevant geographic unit for measuring information spillovers.⁶

The first wave was implemented from April to July 2022. In medium-saturation tehsils, the assigned treatment probability was either 0.35 or 0.45; in high-saturation tehsils, it was either 0.55 or 0.65, randomly assigned.⁷ Low-saturation tehsils were assigned lower treatment probabilities, but introducing cluster-level variation so that average treatment probabilities was 11%, but variation in treatment probabilities across clusters was substantial ranging from 0.046 to 0.218.⁸

The second wave was implemented in August and expanded forecast coverage in all saturation arms. This expansion was conducted as a top-up randomization, in the sense that farmers already assigned to receive forecasts remained treated, and additional farmers were randomly selected from those not yet assigned to treatment. The low saturation group was assigned to 0.25 treatment probabilities on average, and the middle and high saturation groups were assigned to 0.6 and 0.8 treatment probabilities on average, respectively.

The second-wave expansion also incorporated prior engagement with PxD calls, as we introduced 5% local perturbation on treatment probabilities for high and low engaged users, classified as high-engaged if their response rate to calls was above the median response rate. This small additional perturbation was meant to learn marginal effects on these two specific subgroups, but, because unpowered, will be omitted throughout our analysis.

In summary, our main analysis uses the randomized variation in individual assignment and tehsil-level saturation generated by this design. For outcomes measured during the agricultural season, treatment status and tehsil saturation are defined according to the assignment in place at the time of the survey. For endline outcomes, which summarize production over the full season, we define treatment as an indicator for whether the farmer was ever assigned to receive weather

⁶Tehsils are large enough that information may not diffuse immediately to all farmers, but small enough that farmers may plausibly interact directly or indirectly within the same administrative unit.

⁷These variations allowed us to further learn marginal effects as we describe in our theoretical analysis in [Viviano and Rudder \(2024\)](#).

⁸Specifically, for low-saturation tehsils we stratified treatment assignments by the number of active PxD users across tehsils, generating low-saturation probabilities ranging from 0.046 to 0.218 and providing us with additional variation that we can use in our parametric specifications.

forecasts during the 2022 season and define saturation using the final assigned treatment share in the farmer’s tehsil after the second-wave expansion. We interpret estimates as intention-to-treat effects of being assigned access to localized forecasts.

Finally, because the second wave expanded coverage later in the season, some treated farmers had shorter exposure to the forecasts. We therefore report robustness checks that exclude farmers first assigned to treatment in the second wave.

3.2 Data and Sample

We leverage three data sources in our analysis: administrative data the records whether participants listen to phone-calls, a phone survey throughout the 2022 cotton-growing season, and an endline survey with a subset of participants 8 months after the end of the project in July 2023.

3.2.1 Administrative data

Administrative data provided by the NGO provide information on all phone numbers provided by the government roster. It includes records for dates when messages were delivered, the length of the phone message, and the length of the phone message listened to by the participant for both weather forecasts and agronomic advisory. Administrative data are available for both the pre-treatment season which took place from May to October, 2021, and during the project season which took place from May to October, 2022.

3.2.2 2022 Mid-season phone survey

During the 2022 agricultural season, we implemented a phone survey with a sample of farmers drawn from the administrative data records. The primary goal of the survey was to collect real-time data on weather responses from different farmers throughout the season. The primary outcomes collected during the phone survey was to elicit weather predictions for the next day, recall information for weather in the prior day, the exact days of recent farming activities (irrigation, fertilizer, pesticide applications), information about sharing weather, and expectations for yields during harvest. The survey firm had a capacity to call up to 400 farmers per week to complete a survey. We collected a repeated cross section of new farmers each week to maximize geographic coverage. At the end of the season, we have surveys for 5,700 farmers, of which 3,600 were cotton-growers. For most analysis we use the full sample of farmers, but we only asked details on agricultural inputs for cotton-growers.

For weekly phone surveys, we drew random lists of farmer phone numbers to be approximately 50% treatment and 50% control in order to optimize power. We also stratified call lists based on tehsil locations to ensure that approximately equal numbers of users were called from each tehsil for each weekly survey. Over the course of the season, the survey company attempted to call 17,600 users, of which 6,500 completed a survey (5,700 for our sample), resulting in a survey completion rate of 37%. Although this is low, it is in line with phone survey response rates in some other low- and middle-income settings (Gibson et al., 2019, Henderson et al., 2020; Gourlay et al., 2021). In Appendix Table 3, we show how the survey population differs from the full roster of phone numbers.

3.2.3 2023 Endline phone survey

Approximately eight months after the mid-season phone survey, we collected another phone survey with 2,000 respondents. These respondents were a randomly drawn subsample from the original phone survey, balanced on treatment status and location. We asked a series of follow-up questions related to cotton productivity, including yields, profits, costs, and timing of harvest. We also asked farmers about their planting behavior in the subsequent season.

3.3 Main Outcomes

We pre-registered the following outcomes: engagement with weather and advisory phone calls, weather forecast predictions, sharing of information, timing of input applications, yields, and input use and profits. In addition, we pre-registered short and long-run expectations and beliefs and pre-specified forecast accuracy and weather trends (shocks and extreme weather events) as a source of heterogeneity. Before the second survey, we registered short-run adaptation (crop, seed choices, conservation agriculture practices) and perceptions of climate change as additional outcomes.

Forecast predictions and recall: Messages of forecasts sent to mobile phones included an estimate of the expected amount of rain in millimeters, the maximum temperature, and the minimum temperature in celsius. Forecasts did not include probabilistic information. Instead the forecast provider worked with the NGO to establish thresholds that would define whether forecasts would positively report rain levels. During the mid-season phone survey, farmers were asked their predictions for weather for the day following the survey. For rainfall, farmers were asked “Do you think it will rain tomorrow?” For temperature, farmers were asked, “What do you think the max(min) temperature will be tomorrow?” We then compared farmers’ predictions with the forecast information provided on those days. For rainfall, we define a binary variable equal to one

if the farmer’s rain prediction aligns with the forecast (e.g. if a farmer predicts any rain level above zero and the forecast also reports any rain level above zero, they are assigned one; zero otherwise). For temperature, we calculated the absolute deviation of predicted temperature minus the forecast temperature so that predictions above and below the forecast are treated similarly. Negative values in the temperature prediction indicate that the farmer’s prediction was closer to the forecast.

Sharing information: The first variable in this outcome grouping is a binary variable equal to one if farmers reported learning weather information from any official source over the prior week. Second, farmers were asked if they shared weather information with any peers. Finally, farmers were asked if any of their peers shared weather information with them.

Farming behavior: During each mid-season phone survey, farmers were asked if they carried out specific farming tasks and the precise date of those tasks. The primary tasks are planting, irrigation, fertilizing, adding pesticides, herbicides, and fungicides. Not all farmers completed all tasks during a given week; as a result, sample sizes vary based on whether a task was reported during the phone survey. We then compared the dates of tasks with both the realized and forecast weather for the same day. For the main results, we report results for realized weather, and report results for forecasts in the appendix.

3.4 Balance of surveyed participants

We report three sets of balance checks. Table 1 compares phone-survey respondents by individual treatment assignment. Most baseline characteristics are similar across treatment and control respondents, including age, education, household size, land ownership, farm size, prior-season yields, and pre-treatment engagement with PxD advisory calls. The only exception is the share of women, which is statistically different: women account for 4.2% of control respondents and 3.1% of treated respondents. This difference is statistically significant but small in absolute terms, and women represent a small share of the surveyed sample overall.

Table 2 compares respondents across low- and higher-saturation tehsils. These comparisons are reassuring overall. Gender, age, education, land ownership, farm size, prior yields, smartphone use, WhatsApp use, and prior advisory listening are similar across saturation arms. The main statistically significant difference in farmer characteristics is household size: respondents in low-saturation tehsils report about 6.1 household members, compared with 5.3 in higher-saturation tehsils. This difference is modest relative to the dispersion in household size and is not directly related to the main weather-information or input-timing outcomes. We also observe statistically significant differences

in minimum-temperature stress and minimum-temperature forecast accuracy. These differences are less central for our main results, which focus primarily on rainfall and maximum-temperature heat margins; importantly, saturation arms are balanced on cumulative rainfall, days with rain, heavy rain, extreme rain, and forecast accuracy for rainfall and maximum temperature.

Table 3 addresses a different issue: representativeness of the phone-survey sample relative to the full administrative roster. Survey respondents are similar to the administrative population on several demographics, but they are more engaged with the PxD platform before treatment: they have higher prior advisory listening rates and are more likely to have picked up advisory and weather calls. They are also somewhat more educated, more likely to own land, have slightly larger farms, and are more likely to use WhatsApp. These differences are expected in a phone survey of users of a voice-call advisory service, since farmers who answer and listen to calls are also more likely to complete surveys. We therefore interpret survey-based estimates as effects for the reachable phone-survey population, while using administrative call records to characterize engagement in the broader roster when possible.

4 Estimation

4.1 Intention to Treat and Total Spillover

To estimate intention to treat and total spillover effects, we use the following specification:

$$Y_{ic} = \alpha + \beta_1 \text{Treat}_{ic} + \beta_2 \text{Saturation}_c + \epsilon_{ic} \quad (1)$$

Y_{ic} denotes the outcome for farmer i in cluster c . Standard errors are clustered at the tehsil-level. Here, β_1 is the intent-to-treat effect for individuals assigned to treatment. β_2 captures the total spillover effect, and is composed of spillovers for treated and untreated units. In other words, it includes both the additional effect of being treated along with many treated peers, and the spillover effect of untreated units that have many treated peers (Hudgens and Halloran, 2008; Baird et al., 2018). In this specification, treatment saturation is specified as a continuous variable that varies at the cluster level.⁹

⁹The two wave assignment strategy generates differences in treatment duration for individuals in the sample. In our main analysis, we do differentiate between the two assignment waves. In practice, wave 2 was implemented relatively late in the season. Results are similar when dropping those who received treatment later in the season.

4.2 Spillover Effects on Treated and Untreated Units

To disentangle the spillover effects on treated and untreated units, we use the following saturated regression:

$$Y_{ic} = \alpha + \beta_1 \textit{Treat} \times \textit{LowSaturation}_{ic} + \beta_2 \textit{Control} \times \textit{HighSaturation}_c + \beta_3 \textit{Treat} \times \textit{HighSaturation}_c + \epsilon_{ic} \quad (2)$$

For ease of interpretation, we combine the medium and high saturation groups into one group (called High) to compare against the Low Saturation group. The reference category is control units in the low saturation clusters (we have no pure control clusters). β_1 identifies the effect on treated units in low saturation clusters, β_2 identifies the spillover effect on untreated control units located in high saturation clusters, and β_3 identifies the treatment effect of directly treated units in high saturation clusters. The difference between β_1 and β_3 represents the spillover on treated units.

5 Results

5.1 Weather Predictions

Columns 1, 3, and 5 in Table 4 reports treatment effects regressions following specifications in equations 1 and columns 2, 4, and 6 report results using equation 2. There are three measures based on whether the farmer prediction aligns with forecasts - rainfall, maximum temperature, and minimum temperature, as described in Section 3.3. Forecast recipients who are directly treated have rainfall predictions that are 3.4 percentage points more likely to be aligned with forecasts. The total spillover effect exhibits a positive and significant relationship, a 1% increase in the tehsil share treated increases the probability that farmer predictions align with forecasts by 10.2 percentage points. Results are similar for predictions of maximum and minimum temperature - directly treated units have more aligned predictions and the spillover effect is larger in magnitude. Results from the saturated regressions in columns 2, 4, and 6 suggest that the spillover on treated units has the largest effect, compared to the treated units in low saturation clusters and control units in high saturation clusters.

Non-response for some weather forecast variables was high for maximum and minimum temperature. Only about one quarter of phone survey participants supplied a response in degrees Celsius

that could be converted into a measure of absolute deviation. We think this is because without access to a granular forecasts, people have fewer tools to predict temperature with precision. For rainfall, the question was simpler to answer. Participants were asked a yes or no question, "do you think it will rain tomorrow?" which many more respondents were willing to answer. Table 10 reports results on the likelihood of answering these questions. We see that treated farmers were more likely to provide answers to these questions (columns 1 and 5). As an alternative specification, we impute group-level means and re-run regressions with the full sample (columns 3 and 7) to increase precision in estimates from the underrepresented control group and see that results are similar. While the differential non-response is a concern, we highlight these results to show evidence that treated farmers attended to the information and updated their beliefs.

5.2 Receiving and Sharing Weather Information

Table 5 reports outcomes related to receiving and sharing weather information. In this setting, it is possible that control respondents have access to weather information from other sources besides the forecasts provided by the collaborator. Column 1 reports the probability that a respondent reports learning a weather forecast from an official source in the previous week. About 51% of the control group received forecasts, and treated respondents were 12 percentage points more likely to report receiving forecasts. The most common other sources were internet/smartphone, TV, and radio. The intervention induced treated farmers to share weather information with their peers (columns 3 and 4). About 40% of control farmer share weather information and treated farmers 10-13 percentage points more likely to share - a 25% increase. Spillover effects in column 6 are particularly relevant - control farmers in high saturation tehsils were 14% more likely to report receiving weather information from peers, despite not having learned information from official sources (column 2). This validates information sharing as a plausible channel for the spillover effects observed in farming behavior described in the subsequent section.

5.3 Short-run Farming Behavior

We focus on two types of farming responses: 1. avoiding rain, and 2. avoiding heat for a series of farming tasks - planting, irrigation, fertilizing, applying pesticides, fungicides, and herbicides, and an index of all 6 tasks. Panel A of Table 7 reports intent-to-treat and total spillovers and Panel B reports spillovers as a saturated regression for avoiding rain using realized weather. Table 7 is organized similarly for avoiding heat using realized weather. In the appendix, Tables 11 and

12 report results using forecasts to establish avoidance behavior and results are similar. Table 6 shows that treated farmers are more likely to avoid rain when irrigating, fertilizing, and applying pesticides, and that there are large spillovers, particularly for irrigation and fertilizing for both control and treated units in high saturation tehsils (columns 2 and 3). Focusing on Panel B, twenty-two percent of control farmers in low saturation tehsils use irrigation on days without rain, treated farmers in low-saturation tehsils are 14% more likely to irrigate on days without rain, and control and treated units in high saturation tehsils are respectively 47% and 68% more likely irrigate on days without rain (point estimates in percentage points are .032, .104, and .151, respectively). Most farmers in this sample use a combination of public and private sources of irrigation, so that this type of reallocation is feasible without renegotiating irrigation time with neighbors on a shared system.

For fertilizing, control farmers in high saturation areas are 30% more likely, and treated respondents in high saturation are 40% more likely to avoid rain.¹⁰ Recall that in Table 2, we show that tehsils in low and high saturation are balanced in terms of seasonal rainfall patterns and rainfall accuracy, which reduces concerns that highly saturated tehsils systematically differed in terms of rainfall patterns. We see similar patterns for pesticide applications and for the index in column 7. Fungicide and herbicide exhibit noisier and less clear overall patterns, but fewer farmers applied these inputs overall. It appears that treated farmers were less likely to avoid rain when planting, but this was early in the season after farmers only had been receiving forecasts for a few weeks. If farmers planted before treated began, they are excluded, which is why the sample size for planting is lower than for irrigation or fertilizing.

Table 7 reports analogous results for avoiding heat, defined as days with maximum temperature above 37C. Results are statistically weaker, but exhibit similar patterns across all tasks: treated farmers in low and high saturation tehsils as well as control units in high saturation tehsils all avoid heat to a greater extent than control units in low saturation tehsils. One major difference is that control units in low saturation tehsils are ‘better’ at avoiding heat compared to avoiding rain: the share of control farmers that avoid heat ranges from 81-91% for most tasks, compared to 20-30% for avoiding rain. Nonetheless, treated farmers in high saturation areas are 9% more likely to avoid heat when irrigating, 8% more likely during fertilizing, and 12% more likely to avoid heat when applying fungicides.

¹⁰Some types of common fertilizer are water soluble and benefit from some rain to dissolve. We lack precise information about the types of fertilizers used to inspect this in detail but argue that farmers would still prefer to apply fertilizers on days without rain and use irrigation to dissolve fertilizers as needed.

5.4 Productivity Outcomes

Panel A of Table 8 reports spillover effects for productivity outcomes, including yield, profits, revenue, input costs, output prices, and harvest timing. Column 1 shows that yields for treated farmers are 5% higher for all treated farmers and 3-7% larger in saturated regressions. Results are only marginally significant for the treated farmers in low saturation tehsils (Panel B, column 1). In both panels, most measures are too noisy to reject differences from zero, but patterns suggest that profit decreases as saturation rates increase and revenue slightly increases for treated farmers, but not spillover farmers. In both Panel A and B, input costs rise as saturation rates increase, and is statistically larger for treated farmers in high saturation tehsils whose expenditure increased by 12%.

Columns 5 and 6 report the output prices for cotton and the harvest timing. Although measured with noise, we see that prices are 1.6-3.8% lower in higher saturation tehsils, suggesting that one reason why profits are lower is that farmers received lower farmgate prices when selling cotton. Treated farmers in low saturation tehsils harvest their crop later by a few weeks - control farmers harvest in late September and treated farmers in low saturation tehsils harvest in early October. These patterns suggest that there may high saturation may lead to some crowding out at the time of harvest. Harvest is labor-intensive and 90% of farmers hire labor during the period. Treated farmers in low saturation tehsils may have had an information advantage to delay harvests which afforded slightly better prices and overall better profits.

In Panel B, input costs rose by 9-13%, which was statistically different than zero. Columns 5 and 6 show suggestive evidence that treated farmers harvested later and received lower prices, which is a possible mechanism to explain lower overall profits. Panel B reports input costs per acre and shows that costs increased by 9-16% for irrigation, pesticides, and fertilizer. Given that farmers responded to treatment by adjusting the timing of their input applications, it suggests that some of these adjustments were costly in the short-run.

5.5 Forecast Accuracy and Take-up

Among participants in the 2022 phone survey, 75% listened to at least one weather message and 77% listened to at least one advisory message. On average, people who listened to at least one weather message listened to 30% of all messages sent, highlighting that people selectively answered messages throughout the season. Table 9 reports differences in message take-up rates by different measures of

forecast accuracy. First, satellite rainfall accuracy (row 1) refers to the share of days where rainfall forecasts aligned with realized rainfall according to satellite measures. Farmer-reported rainfall accuracy (2) is the share of days when whether forecasts aligned with farmer-reported rainfall during the survey. Absolute rainfall deviations (3) is the average absolute deviation of forecasts from realized weather, measured in millimeters. Rainfall false alarms (4) is the number of days with false alarms in precipitation, defined as days where forecasts predicted no rain and a heavy rain fell, or when forecasts predicted heavy rain and no rain fell. Rainfall RMSE (5), max temp (6), and min temp (7) are the root mean-squared error for rainfall, max and min temp from a tehsil-level regression of the realized value on the forecast value. Measures 3-7 are common measures used in meteorology to assess forecast accuracy.

Measures 1 and 2 were constructed as binary measures that can be interpreted as the share of days where a forecast of positive rain corresponded to realized rain, and vice versa for forecasts and realized weather without rain. Measure 1 uses satellite measures of realized rain compared to forecasts. Measure 2 uses farmer-reported rainfall. During the phone survey, respondents were asked if it rained the prior day and we matched that up with forecast on the same day. Measure 2 is the 'ground truth' data that best captures whether forecasts were accurate for participants. According to both measures, increased accuracy is correlated with higher average take-up, suggesting that farmers in areas with lower accuracy decreased their take-up as a response. A 10% increase in the share of days with an accurate forecast leads to a 1.6-2.9% increase in average take-up. A few of the other satellite-based measures of accuracy show the opposite trend; an increase in rainfall deviations between forecast and realized weather (row 3) and an increase in max temp RMSE (row 6) are related to a an increase in take-up (by construction, higher values of measures 3-7 imply *lower* accuracy). However, these results are not necessarily inconsistent, measures 1 and 2 are binary measures while 3 and 6 are continuous. It is possible that higher rainfall deviations correspond to days with rain, and the binary measure of 'yes' or 'no' rain are what matters. For temperature, overall RMSE was low across tehsils and in general temperature forecasts tend have higher accuracy because they are more spatially correlated compared to rainfall.

6 Conclusion

After implementing a saturation design at large-scale with 400,000 farmers in 35 tehsils, we show that information spillovers are common and influence farmer decision making. We separately

identify intent-to-treat effects, total spillovers, as well as spillovers on treated and untreated farmers. We show a pattern of behavior change that indicates that spillovers occur for both treated and control farmers, indicating that treated farmers benefit from having other treated farmers to discuss farming and that control farmers benefit indirectly through information sharing.

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7 Tables and Figures

Experimental Balance

Table 1: Experimental Balance by Individual Treatment Status

Variable	(1) Control Mean/SD	(2) Treatment Mean/SD	T-test Difference Control-Treat
Woman	0.042 (0.201)	0.031 (0.186)	0.011***
Age	42.100 (16.420)	41.914 (14.841)	0.186
Education	5.970 (7.086)	5.876 (8.189)	0.094
Household size	5.464 (4.022)	5.413 (3.284)	0.051
Owns land	0.868 (0.654)	0.877 (0.613)	-0.010
Farm size (acres)	5.578 (5.925)	5.522 (6.454)	0.055
Uses smartphone (self-reported)	0.342 (0.616)	0.354 (0.434)	-0.012
Uses WhatsApp (automated)	0.322 (0.464)	0.331 (0.619)	-0.009
Prior season yields (mounds/acre)	25.938 (13.403)	25.666 (15.319)	0.272
Prior season advisory listening rate on all calls	0.566 (0.675)	0.580 (0.725)	-0.015
Prior season advisory listen at least once	0.936 (0.286)	0.942 (0.291)	-0.006
Completed Farming Task Module	0.611 (1.023)	0.604 (1.085)	0.007
N	2547	3187	
Clusters	35	35	
F-test of joint significance (F-stat)			3.479***
F-test, number of observations			5734

Table 2: Experimental Balance by Cluster Saturation

Variable	(1) Low Saturation Mean/SD	(2) Medium and High Saturation Mean/SD	T-test Difference Low-High
Woman	0.036 (0.253)	0.036 (0.208)	-0.000
Age	41.512 (16.997)	42.046 (15.271)	-0.534
Education	5.735 (8.530)	5.949 (9.692)	-0.214
Household size	6.136 (6.791)	5.294 (3.966)	0.842***
Owns land	0.883 (0.474)	0.871 (0.797)	0.012
Farm size (acres)	5.207 (9.518)	5.611 (7.915)	-0.404
Uses smartphone (self-reported)	0.335 (0.318)	0.350 (0.512)	-0.016
Uses WhatsApp (automated)	0.312 (0.509)	0.329 (0.588)	-0.017
Prior season yields (mounds/acre)	25.683 (15.714)	25.803 (16.484)	-0.120
Prior season advisory listening rate on all calls	0.558 (0.688)	0.576 (0.841)	-0.018
Prior season advisory listen at least once	0.942 (0.270)	0.939 (0.330)	0.003
Completed Farming Task Module	0.587 (1.427)	0.609 (1.313)	-0.022
Days with max temp stress	67.421 (423.113)	49.711 (548.746)	17.710
Days with min temp stress	0.148 (6.337)	16.357 (426.056)	-16.209**
Days with max temp extreme	7.710 (139.385)	5.793 (137.749)	1.917
Days with min temp extreme	0.000 (0.000)	4.430 (205.925)	-4.430
Cumulative Rainfall	686.679 (682.429)	691.990 (1151.743)	-5.311
Days with extreme rain	18.938 (27.176)	18.701 (41.437)	0.237
Days with any rain	110.879 (130.830)	108.540 (219.532)	2.339
Days with heavy rain	28.896 (49.395)	27.355 (54.634)	1.541
Days with no rain	72.121 (130.830)	74.460 (219.532)	-2.339
Forecast accuracy - rain	8.528 (10.688)	9.118 (19.156)	-0.590
Forecast accuracy - max temp	1.742 (5.314)	1.870 (3.421)	-0.128
Forecast accuracy - min temp	1.130 (2.355)	1.447 (5.665)	-0.317***
N	1608	4126	
Clusters	10	25	
F-test of joint significance (F-stat)			0.696
F-test, number of observations			5734

Table 3: Difference in Means for Administrative Data and Phone Survey Sample

	Administrative Data	Completed Phone Survey	
	Mean (sd)	Mean (sd)	Test
Woman	0.043 (0.204)	0.034 (0.182)	0.015
Age	51.132 (14.455)	50.918 (13.644)	0.419
Education	5.457 (4.309)	5.949 (4.237)	<0.001
Household size	5.404 (3.303)	5.506 (3.177)	0.092
Owns land	0.852 (0.355)	0.872 (0.334)	0.002
Farm size (acres)	5.193 (4.480)	5.508 (4.718)	<0.001
Uses WhatsApp (automated)	0.267 (0.443)	0.323 (0.467)	<0.001
Prior season advisory listening rate on all calls	0.426 (0.346)	0.575 (0.317)	<0.001
Prior season advisory listen at least once	0.786 (0.410)	0.939 (0.239)	<0.001
Weather listening rate	0.125 (0.223)	0.185 (0.262)	<0.001
Pick up weather at least once	0.438 (0.496)	0.512 (0.500)	<0.001
Advisory listening rate	0.208 (0.253)	0.327 (0.274)	<0.001
Pick up advisory at least once	0.774 (0.418)	0.958 (0.200)	<0.001
Treat	0.611 (0.488)	0.556 (0.497)	<0.001
N	442,783 (98.7%)	5,734 (1.3%)	

Main Results Tables

Table 4: ITT and Spillover Effects of Weather Predictions

	Rainfall		Max Temperature		Min Temperature	
	(1) Prediction equals forecast (0/1)	(2) Prediction equals forecast (0/1)	(3) Absolute Deviation from Forecast	(4) Absolute Deviation from Forecast	(5) Absolute Deviation from Forecast	(6) Absolute Deviation from Forecast
Treat	0.034*** (0.012)		-0.676*** (0.172)		-1.074*** (0.232)	
Tehsil Share Treated	0.102*** (0.035)		-1.026** (0.426)		-1.033** (0.412)	
Treat×Low Saturation		0.023 (0.023)		-0.486 (0.475)		-1.499*** (0.493)
Control×High Saturation		0.008 (0.027)		-0.224 (0.489)		-0.698* (0.381)
Treat×High Saturation		0.051* (0.025)		-1.003** (0.471)		-1.720*** (0.310)
Pvalue: Treat(Low)=Control(High)		0.5188		0.3728		0.0839
Pvalue: Treat(Low)=Treat(High)		0.2682		0.0850		0.6005
Pvalue: Control(High)=Treat(High)		0.0086		0.0002		0.0004
Control Mean	0.438	0.432	4.056	4.227	4.915	5.446
Observations	5144	5144	1283	1283	1189	1189
Adj R-Squared	0.004	0.001	0.019	0.015	0.026	0.024

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent's predicted forecast for rainfall, maximum temperature, and minimum temperature for the following day align with the forecasts provided by phone messages. Sample sizes differ due to non-response. Results are similar if group means are imputed as reported in Table 10 in the appendix.

Table 5: ITT and Spillover Effects of Receiving and Sharing Weather Information

	(1) Received Weather Info from Official Sources	(2) Received Weather Info from Official Sources	(3) Shared Weather Info w/ Peers	(4) Shared Weather Info w/ Peers	(5) Peers Shared Weather Info	(6) Peers Shared Weather Info
Treat	0.119*** (0.014)		0.105*** (0.012)		0.029* (0.015)	
Tehsil Share Treated	-0.035 (0.044)		-0.006 (0.041)		-0.015 (0.033)	
Treat×Low Saturation		0.108*** (0.034)		0.131*** (0.028)		0.113*** (0.025)
Control×High Saturation		-0.010 (0.028)		0.015 (0.025)		0.048** (0.019)
Treat×High Saturation		0.109*** (0.028)		0.113*** (0.026)		0.056*** (0.017)
Pval: T(Low)=C(High)		0.0034		0.0004		0.0111
Pval: T(Low)=T(High)		0.9710		0.5405		0.0082
Pval: C(High)=T(High)		0.0000		0.0000		0.5738
Control Mean	0.509	0.516	0.399	0.387	0.395	0.358
Observations	5733	5733	5732	5732	5723	5723
Adj R-Squared	0.014	0.013	0.011	0.011	0.001	0.002

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent received weather information from an official source such as government, TV, Radio, Mobile phone (Columns 1 and 2), whether they shared weather information with peers (Columns 3 and 4), and whether peers shared weather information with the respondent over the prior week (Columns 5 and 6). Columns 1, 3, and 5 report ITT treatment effects and total spillover effects as a continuous measure equal to the share of units treated in each tehsil. Columns 2, 4, and 6 report saturated regression where the reference category is control units in low saturation clusters. All outcomes are binary.

Table 6: ITT and Spillover Effects on Farming Behavior: Avoiding Rain

Panel A: ITT and Spillover Effects as Continuous Measure							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Rain Index
Treat	-0.036* (0.020)	0.038** (0.015)	0.010 (0.015)	0.029* (0.016)	0.005 (0.027)	-0.011 (0.021)	0.003 (0.017)
Tehsil Share Treated	-0.062 (0.108)	0.247*** (0.081)	0.127* (0.064)	0.126* (0.067)	-0.047 (0.078)	0.015 (0.083)	0.051 (0.075)
Control Mean	0.552	0.256	0.190	0.235	0.259	0.264	0.000
Observations	2257	3303	3223	3046	1167	2071	3482
Adj R-Squared	0.001	0.019	0.005	0.006	-0.001	-0.001	0.000
Panel B: Spillover Effects as Saturated Regression							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Rain Index
Treat×Low Saturation	-0.015 (0.020)	0.032* (0.017)	0.005 (0.018)	0.022 (0.021)	0.026 (0.036)	-0.024 (0.023)	0.005 (0.017)
Control×High Saturation	0.067 (0.064)	0.104*** (0.034)	0.053** (0.023)	0.051 (0.031)	-0.023 (0.038)	-0.011 (0.047)	0.045 (0.038)
Treat×High Saturation	-0.064* (0.036)	0.151*** (0.027)	0.070** (0.029)	0.087** (0.033)	-0.044 (0.028)	0.007 (0.039)	0.041 (0.042)
Pval: T(Low)=C(High)	0.2339	0.0534	0.0932	0.3557	0.3129	0.7783	0.3160
Pval: T(Low)=T(High)	0.2382	0.0002	0.0467	0.0507	0.0949	0.4316	0.4083
Pval: C(High)=T(High)	0.0064	0.0728	0.4575	0.1239	0.5721	0.6963	0.9403
Control Mean	0.540	0.223	0.174	0.217	0.267	0.267	0.000
Observations	2257	3303	3223	3046	1167	2071	3482
Adj R-Squared	0.002	0.017	0.005	0.005	0.001	-0.001	0.000

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent carried out farming tasks on days without rain according to satellite measures of realized weather. Panel A reports ITT effects as a binary variable and spillover effects as a continuous measure set to the saturation level of each tehsil. Panel B reports saturated regression with the reference category of control units in low saturation tehsils. Panel B also reports t-tests of equality for each saturation by treatment pair. Sample sizes differ because responses are conditional on having reported a date for each task.

Table 7: ITT and Spillover Effects on Farming Behavior: Avoiding Heat

Panel A: ITT and Spillover Effects as Continuous Measure							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Heat Index
Treat	0.009 (0.020)	0.024** (0.012)	0.016 (0.010)	0.020** (0.009)	0.029 (0.024)	0.045*** (0.015)	0.036** (0.013)
Tehsil Share Treated	0.220 (0.289)	0.125 (0.090)	0.106 (0.082)	0.072 (0.082)	0.171* (0.094)	0.165 (0.114)	0.125 (0.135)
Control Mean	0.437	0.865	0.880	0.918	0.838	0.771	0.000
Observations	2293	3303	3225	3046	1170	2081	3482
Adj R-Squared	0.007	0.010	0.007	0.006	0.014	0.011	0.006
Panel B: Spillover Effects as Saturated Regression							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Heat Index
Treat×Low Saturation	-0.007 (0.023)	0.039** (0.016)	0.015 (0.014)	0.041*** (0.013)	0.037 (0.034)	0.047** (0.017)	0.040** (0.018)
Control×High Saturation	0.059 (0.166)	0.078* (0.042)	0.053 (0.040)	0.028 (0.038)	0.075 (0.051)	0.056 (0.070)	0.054 (0.090)
Treat×High Saturation	0.132 (0.175)	0.078* (0.040)	0.068* (0.036)	0.028 (0.038)	0.100** (0.049)	0.098 (0.064)	0.091 (0.085)
Pval: T(Low)=C(High)	0.6992	0.3494	0.3737	0.7036	0.5053	0.8978	0.8694
Pval: T(Low)=T(High)	0.4387	0.3179	0.1607	0.6990	0.2125	0.4264	0.5378
Pval: C(High)=T(High)	0.0752	0.9985	0.1746	0.9898	0.2290	0.1189	0.2876
Control Mean	0.427	0.841	0.865	0.908	0.813	0.757	0.000
Observations	2293	3303	3225	3046	1170	2081	3482
Adj R-Squared	0.006	0.010	0.007	0.003	0.010	0.006	0.004

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent carried out farming tasks on days without extreme heat above 37C according to satellite measures of realized weather. Panel A reports ITT effects as a binary variable and spillover effects as a continuous measure set to the saturation level of each tehsil. Panel B reports saturated regression with the reference category of control units in low saturation tehsils. Panel B also reports t-tests of equality for each saturation by treatment pair. Sample sizes differ because responses are conditional on having reported a date for each task.

Table 8: Treatment Effects on Farming Productivity

Panel A: ITT and Spillover Effects as Continuous Measure						
	(1) Yields	(2) Profits	(3) Revenue	(4) Input Costs	(5) Output Price	(6) Harvest Month
Treat=1	0.603 (0.592)	1705.830 (4926.933)	4279.998 (5168.249)	2227.718 (1871.689)	21.413 (117.490)	0.038 (0.061)
Tehsil Share Treated	-0.284 (2.540)	-15120.842 (16146.060)	-7830.191 (19157.336)	9887.454 (5893.545)	-124.340 (607.263)	-0.021 (0.156)
Control Mean	11.780	25005.653	94968.054	69699.634	7703.776	9.829
Observations	1403	1386	1386	1408	1386	1323
Adj R-Squared	0.000	0.001	-0.000	0.005	-0.000	-0.002
Panel B: Spillover Effects as a Saturated Regression						
	(1) Yields	(2) Profits	(3) Revenue	(4) Input Costs	(5) Output Price	(6) Harvest Month
Treat in Low	0.815* (0.451)	921.211 (5255.108)	5843.102 (5814.270)	4967.618 (3219.009)	12.674 (265.888)	0.251** (0.119)
Control in High	0.317 (1.456)	-5932.430 (8272.195)	-1496.390 (9523.803)	5724.317 (3804.726)	-298.533 (334.257)	0.118 (0.093)
Treat in High	0.796 (1.420)	-4455.690 (7399.278)	2313.280 (8798.241)	7626.130** (3721.279)	-190.940 (344.409)	0.085 (0.089)
Pval: T(Low)=C(High)	0.7265	0.3069	0.4852	0.8630	0.2915	0.1287
Pval: T(Low)=T(High)	0.9891	0.3898	0.7161	0.5449	0.5116	0.0695
Pval: C(High)=T(High)	0.4666	0.7832	0.4904	0.3473	0.3843	0.5689
Control Mean	11.504	29564.188	95816.595	64999.259	7922.917	9.741
Observations	1403	1386	1386	1408	1386	1323
Adj R-Squared	-0.001	-0.001	-0.001	0.002	0.001	0.001

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for productivity outcomes. Panel A reports ITT effects as a binary variable and spillover effects as a continuous measure set to the saturation level of each tehsil. Panel B reports saturated regression with the reference category of control units in low saturation tehsils. Panel B also reports t-tests of equality for each saturation by treatment pair.

Table 9: Take-up and Forecast Accuracy

	Weather Message Take-up
(1) Satellite Rainfall Accuracy	0.288** (0.119)
(2) Farmer-Reported Rainfall Accuracy	0.155* (0.087)
(3) Satellite Absolute Rainfall Deviation	0.025* (0.013)
(4) Rainfall False Alarms	-0.001 (0.003)
(5) Rainfall RMSE	0.004 (0.005)
(6) Max Temp RMSE	0.041** (0.017)
(7) Min Temp RMSE	0.013 (0.015)
Take-up Mean	0.217
Observations	4964

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports separate regressions for the relationship between measures of accuracy (rows 1-7) and average take-up rates of weather forecast messages. The unit of analysis is tehsil by day. There are 35 unique tehsils and each received between 140-145 weather messages throughout the season. (1) refers to the share of days where rainfall forecasts aligned with realized rainfall according to satellite measures. (2) is the share of days when whether forecasts aligned with farmer-reported rainfall during the survey. (3) is the average absolute deviation of forecasts from realized weather, measured in millimeters. (4) is the number of days with false alarms in precipitation, defined as days where forecasts predicted no rain and a heavy rain fell, or when forecasts predicted heavy rain and no rain fell. (5), (6), and (7) are the root mean-squared error for rainfall, max temp, and min temp from a tehsil-level regression of the realized value on the forecast value.

Appendix

Table 10: Response Rates for Maximum and Minimum Temperature Weather Predictions

	Maximum Temperature				Minimum Temperature			
	(1) Responds to Max Temp Forecast (0/1)	(2) Responds to Max Temp Forecast (0/1)	(3) Absolute Deviation from Forecast	(4) Absolute Deviation from Forecast	(5) Responds to Min Temp Forecast (0/1)	(6) Responds to Min Temp Forecast (0/1)	(7) Absolute Deviation from Forecast	(8) Absolute Deviation from Forecast
Treat	0.046*** (0.013)		-1.507*** (0.046)		0.043*** (0.013)		0.019 (0.047)	
Tehsil Share Treated	-0.045 (0.034)		-0.212** (0.103)		-0.051 (0.036)		-0.271*** (0.081)	
Treat in Low		0.012 (0.028)		-1.442*** (0.106)		0.005 (0.026)		-0.078 (0.127)
Control in High		-0.014 (0.029)		-0.040 (0.107)		-0.017 (0.029)		-0.172** (0.083)
Treat in High		0.036 (0.026)		-1.570*** (0.107)		0.031 (0.028)		-0.140* (0.070)
Pval: T(Low)=C(High)		0.2632		0.0000		0.3395		0.3612
Pval: Treat(Low)=T(High)		0.2822		0.0787		0.2717		0.5412
Pval: C(High)=T(High)		0.0036		0.0000		0.0035		0.5428
Control Mean	0.438	0.432	4.056	4.227	0.438	0.432	4.915	5.446
Observations	5144	5144	5144	5144	5144	5144	5144	5144
Adj R-Squared	0.003	0.002	0.192	0.192	0.003	0.002	0.001	0.000

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT treatment effects regressions for robustness on weather predictions. Columns 1-2 and 5-6 report whether a respondent answered the survey question that asked for a prediction for maximum and minimum temperature for the following day. Columns 1 and 5 show that treated farmers are more likely to provide temperature predictions, but results are noisier and smaller in magnitude in columns 2 and 6, showing that control farmers in high saturation tehsils had similar response rates. Columns 3-4 and 7-8 report prediction errors for maximum and minimum temperature after imputing group means for non-response.

Table 11: ITT and Spillover Effects on Farming Behavior: Avoiding Forecasted Rain

Panel A: ITT and Spillover Effects as Continuous Measure							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Rain Index
Treat	-0.037 (0.089)	0.051** (0.020)	0.025 (0.017)	0.035** (0.014)	-0.003 (0.027)	-0.004 (0.016)	-0.004 (0.014)
Tehsil Share Treated	0.064 (0.196)	0.223*** (0.069)	0.108** (0.046)	0.150*** (0.054)	0.079 (0.066)	0.094* (0.053)	0.052 (0.034)
Control Mean	0.628	0.275	0.176	0.225	0.187	0.167	0.000
Observations	137	3213	3082	2932	968	1567	3482
Adj R-Squared	-0.013	0.017	0.005	0.009	0.000	0.002	0.000
Panel B: Spillover Effects as Saturated Regression							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Rain Index
Treat×Low Saturation	0.016 (0.105)	0.053** (0.025)	0.027 (0.016)	0.008 (0.020)	-0.033 (0.031)	-0.012 (0.018)	-0.005 (0.016)
Control×High Saturation	0.144 (0.100)	0.087** (0.034)	0.046* (0.024)	0.054* (0.031)	-0.025 (0.039)	0.020 (0.032)	0.055 (0.039)
Treat×High Saturation	-0.106 (0.116)	0.140*** (0.029)	0.070*** (0.024)	0.116*** (0.030)	0.021 (0.028)	0.033 (0.024)	0.051** (0.023)
Pval: T(Low)=C(High)	0.3191	0.3759	0.5262	0.1030	0.8399	0.3177	0.1270
Pval: T(Low)=T(High)	0.3865	0.0060	0.0916	0.0003	0.0503	0.1301	0.0252
Pval: C(High)=T(High)	0.0488	0.0676	0.4089	0.0005	0.2675	0.7172	0.9300
Control Mean	0.606	0.247	0.163	0.206	0.196	0.162	0.000
Observations	137	3213	3082	2932	968	1567	3482
Adj R-Squared	-0.010	0.012	0.004	0.011	-0.000	-0.000	0.002

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent carried out farming tasks on days without rain according to forecasts provided to farmers. Panel A reports ITT effects as a binary variable and spillover effects as a continuous measure set to the saturation level of each tehsil. Panel B reports saturated regression with the reference category of control units in low saturation tehsils. Panel B also reports t-tests of equality for each saturation by treatment pair. Sample sizes differ because responses are conditional on having reported a date for each task.

Table 12: ITT and Spillover Effects on Farming Behavior: Avoiding Forecasted Heat

Panel A: ITT and Spillover Effects as Continuous Measure							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Heat Index
Treat	-0.005 (0.004)	-0.005 (0.018)	0.012 (0.014)	0.018 (0.014)	0.041 (0.033)	0.059** (0.022)	0.010 (0.019)
Tehsil Share Treated	0.005 (0.012)	0.081 (0.082)	0.135** (0.065)	0.004 (0.061)	0.244*** (0.084)	0.130 (0.111)	0.059 (0.060)
Control Mean	0.013	0.725	0.741	0.800	0.658	0.564	0.000
Observations	2293	3303	3225	3046	1170	2081	3482
Adj R-Squared	-0.000	0.001	0.005	-0.000	0.016	0.006	0.001
Panel B: Spillover Effects as Saturated Regression							
	(1) Planting	(2) Irrigate	(3) Fertilize	(4) Pesticide	(5) Fungicide	(6) Herbicide	(7) Avoid Heat Index
Treat×Low Saturation	-0.007 (0.005)	0.009 (0.021)	0.011 (0.018)	0.057*** (0.016)	0.038 (0.045)	0.065*** (0.024)	0.010 (0.022)
Control×High Saturation	-0.006 (0.006)	0.062* (0.031)	0.042 (0.029)	0.027 (0.028)	0.112** (0.049)	0.032 (0.058)	0.041 (0.043)
Treat×High Saturation	-0.000 (0.010)	0.034 (0.037)	0.062* (0.031)	-0.000 (0.034)	0.164*** (0.040)	0.084 (0.053)	0.053 (0.048)
Pval: T(Low)=C(High)	0.8801	0.1332	0.3403	0.1948	0.2149	0.5879	0.4946
Pval: T(Low)=T(High)	0.5241	0.5112	0.1064	0.0692	0.0108	0.7228	0.3844
Pval: Cont(High)=T(High)	0.4986	0.2515	0.3637	0.1903	0.1529	0.1747	0.6531
Control Mean	0.014	0.705	0.729	0.790	0.621	0.556	0.000
Observations	2293	3303	3225	3046	1170	2081	3482
Adj R-Squared	-0.000	0.002	0.002	0.003	0.017	0.003	0.001

Robust standard errors in parenthesis clustered at the tehsil level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Table reports ITT and spillover treatment effects regressions for whether a respondent carried out farming tasks on days without extreme heat above 37C according to forecasts sent to mobile phones. Panel A reports ITT effects as a binary variable and spillover effects as a continuous measure set to the saturation level of each tehsil. Panel B reports saturated regression with the reference category of control units in low saturation tehsils. Panel B also reports t-tests of equality for each saturation by treatment pair. Sample sizes differ because responses are conditional on having reported a date for each task.